Presented by :
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EduNova:

Pitch Deck
Presentation

EduNova

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Introduction

Project Background -
A curriculum reform meets an access gap

Malaysia's science curriculum has fundamentally shifted toward Higher-Order Thinking Skills (KBAT), moving away from rote memorization to demand deep conceptual understanding and application. This reform relies on a crucial assumption: that students have regular, hands-on experience in science labs to build these mental models. However, severe inequalities in lab access exist, particularly in rural and under-resourced schools. Challenges like missing apparatus, large class sizes, and limited timetable slots mean that for hundreds of thousands of SPM candidates each year, essential practical experiments are reduced to mere teacher demonstrations or textbook diagrams.

This creates a critical national-scale gap right when exam structures demand true comprehension. Fortunately, two forces are converging to provide a solution: the urgent need for accessible practical science, and the maturation of Generative AI. Synapse sits precisely at this intersection. By recreating the interactive moments of physical lab experiments directly on the smartphones that students already own, Synapse ensures that every learner—regardless of their school's funding or physical facilities—can build the hands-on conceptual foundation needed to succeed.

Problem Statement

Incompetency of Traditional Learning Environment

Unequal Access to Crucial Lab Infrastructure

Persistent and Hidden Misconceptions

Inadequate EdTech and AI Solutions

Objectives

1. Deliver Targeted, Pedagogically Sound Interventions

To transform rote-learning approaches by moving beyond generic repetition. The system will dynamically apply precise learning-science strategies—such as contrasting cases for misconceptions or scaffolded examples for knowledge gaps—ensuring students build the true conceptual mastery required by the KBAT curriculum.

2. Provide Universal Access to Faithful Virtual Experiments

To ensure every student, regardless of their school's funding or physical facilities, can experience the hands-on "aha" moments of science. The system will deliver scientifically accurate, interactive virtual labs directly to the personal devices students already own.

3. Diagnose Root Misconceptions with Evidence-Based Precision

To replace surface-level quizzes with a KSSM-aligned diagnostic tool that evaluates student responses against a well-researched database of known scientific misconceptions. This accurately pinpoints specific flawed mental models rather than guessing, allowing for measurable, data-backed correction.

4. Make AI Reasoning Transparent and Auditable

To eliminate the risks of AI hallucinations and "black box" generic tutors. The system will stream its diagnostic and intervention choices live, building metacognitive awareness in students and providing teachers with clear, verifiable oversight of the pedagogical decisions being made.

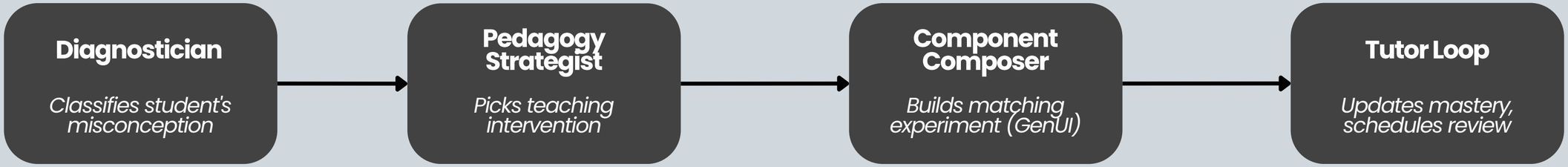
Our Solution

EduNova: The AI tutor that shows its reasoning, not just an answer.

Project Overview

Synapse is a multi-agent AI tutor for Malaysian SPM science students that diagnoses the exact misconception behind a wrong answer, not just "wrong," but which specific wrong idea, then corrects it with a faithful, interactive experiment rather than a generated explanation. It closes the gap left by unequal lab access in B40 and rural schools, giving every student the same hands-on "aha" moment regardless of their school's resources, while showing its reasoning at every step so it's never a black box.

AI Agent Pipeline



Live Demo

Expected Outcomes

KBAT Curriculum Mastery

Transitioning students away from rote memorization and toward the deep conceptual understanding required by Malaysia's science curriculum reform

Universal Lab Access

Delivering scientifically accurate, interactive virtual labs directly to students' smartphones, ensuring equal opportunities for learners in B40 and under-resourced rural schools.

Targeted Misconception Correction

Diagnosing specific flawed mental models rather than just marking answers as "wrong," enabling precise, data-backed educational interventions.

Transparent AI Integration

Eliminating the "black box" effect of standard AI tutors by making pedagogical decisions and reasoning fully visible and auditable for both students and teachers.

Our Final Conclusion

EduNova’s Synapse system effectively bridges the critical gap between Malaysia's ambitious science curriculum and the reality of unequal school infrastructure. By merging the maturation of Generative AI with proven pedagogical strategies, Synapse ensures that the hands-on "aha" moments of scientific discovery are accessible to every SPM student, regardless of their school's funding. Ultimately, EduNova provides a transparent, evidence-based learning environment that democratizes science education and empowers students to achieve true conceptual mastery.

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Thank You

for your time
and attention

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Example

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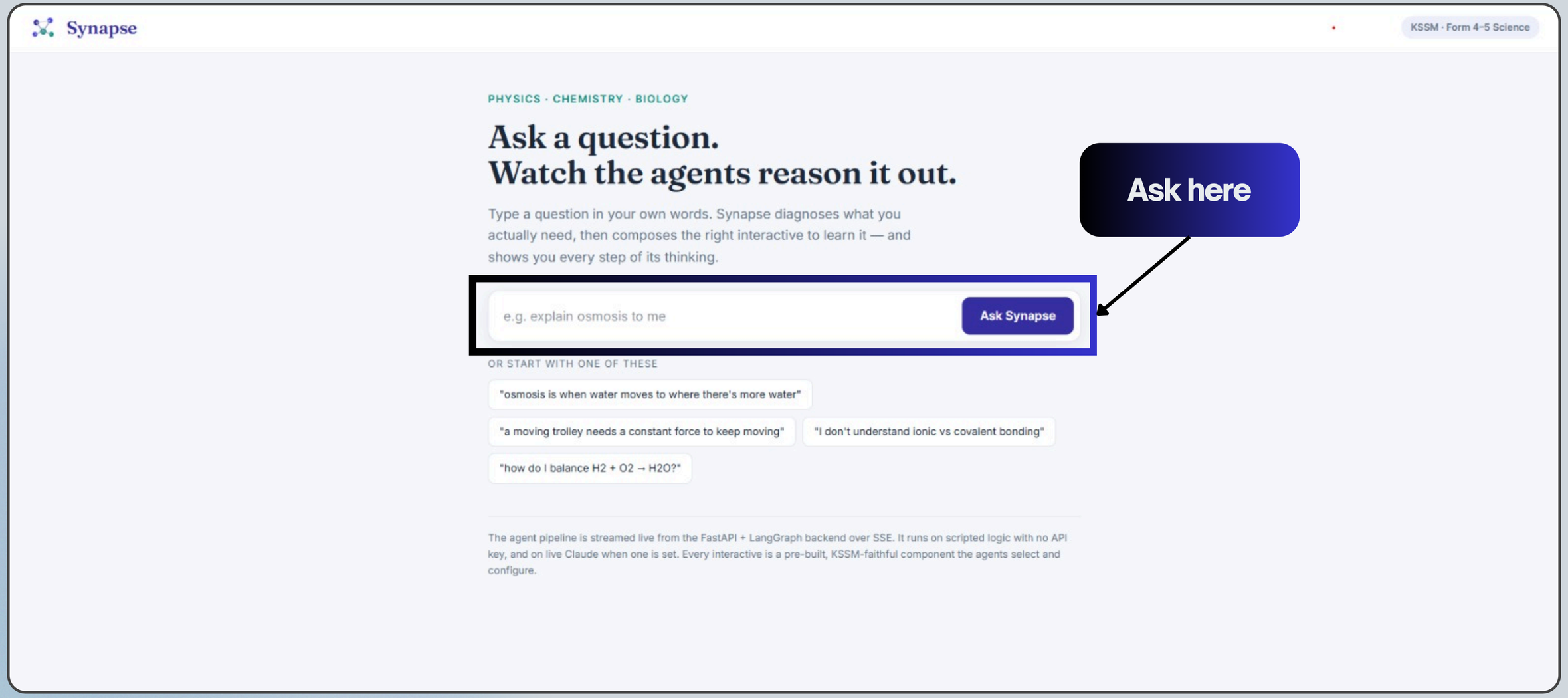


Example

EduNova

Chemistry

I don't understand ionic vs covalent bonding

Ask Synapse

OR START WITH ONE OF THESE

"osmosis is when water moves to where there's more water"

"a moving trolley needs a constant force to keep moving"

"I don't understand ionic vs covalent bonding"

"how do I balance H2 + O2 → H2O?"

REASONING TRACE

1. **Diagnostician** DONE
   Understanding the question...
   Knowledge Gap in ionic vs covalent bonding (Chemistry).
2. **Pedagogy Strategist** DONE
   Choosing the pedagogically-correct move...
   labelled-exploration → compose a "electron-bonding-explorer". Since this is a knowledge gap rather than a misconception, guided labelled exploration of electron transfer versus sharing builds foundational conceptual understanding before contrasting or prediction tasks are needed.
3. **Component Composer** DONE
   Selecting and configuring a "electron-bonding-explorer"...
   Composed a "electron-bonding-explorer" configured for ionic vs covalent bonding.
4. **Tutor Loop** DONE
   Updating the learner profile...
   Noted engagement with ionic vs covalent bonding on the profile.

Bonding & electrons LABELLED-EXPLORATION

Chemistry · Form 4 · ionic vs covalent bonding

Na (metal) ? Cl (non-metal)

example: **NaCl**

Which bond forms? Look at whether each atom is a metal or a non-metal.

Ionic — electrons transfer | Covalent — electrons shared

Reveal the electrons →

Biology

can u explain the menstrual cycle

Ask Synapse

OR START WITH ONE OF THESE

"osmosis is when water moves to where there's more water"

"a moving trolley needs a constant force to keep moving"

"I don't understand ionic vs covalent bonding"

"how do I balance H2 + O2 → H2O?"

REASONING TRACE

1. **Diagnostician** DONE
   Understanding the question...
   Knowledge Gap in menstrual cycle (Biology).
2. **Pedagogy Strategist** DONE
   Choosing the pedagogically-correct move...
   labelled-exploration → compose a "process-timeline". A knowledge gap (not a misconception) about the menstrual cycle's hormonal stages is best addressed by guided labelled exploration of a sequential process to build accurate foundational knowledge.
3. **Component Composer** DONE
   Selecting and configuring a "process-timeline"...
   Composed a "process-timeline" configured for menstrual cycle.
4. **Tutor Loop** DONE
   Updating the learner profile...
   Noted engagement with menstrual cycle on the profile.

28-day menstrual cycle LABELLED-EXPLORATION

Biology · Form 4 · menstrual cycle

Day 1 — Day 8 — Day 28

Follicular phase

FSH matures a follicle; oestrogen rises and rebuilds the lining.

Oestrogen

LH

Progesterone

Physics

can u explain to me snells law

Ask Synapse

OR START WITH ONE OF THESE

"osmosis is when water moves to where there's more water"

"a moving trolley needs a constant force to keep moving"

"I don't understand ionic vs covalent bonding"

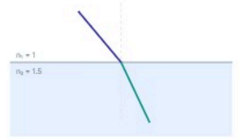
"how do I balance H2 + O2 → H2O?"

REASONING TRACE

1. **Diagnostician** DONE
   Understanding the question...
   Knowledge Gap in Snell's Law (Physics).
2. **Pedagogy Strategist** DONE
   Choosing the pedagogically-correct move...
   labelled-exploration → compose a "wave-optics-sandbox". Since this is a knowledge gap rather than a misconception, a labelled exploration of the wave-optics sandbox helps build foundational understanding of refraction and Snell's Law through guided, structured observation before any prediction-based challenge is needed.
3. **Component Composer** DONE
   Selecting and configuring a "wave-optics-sandbox"...
   Composed a "wave-optics-sandbox" configured for Snell's Law.
4. **Tutor Loop** DONE
   Updating the learner profile...
   Noted engagement with Snell's Law on the profile.

Refraction (Snell's law) LABELLED-EXPLORATION

Physics · Form 5 · Snell's Law



Angle of incidence: **40°**

Entering the denser medium, the ray bends toward the normal: refraction angle ≈ 25.4°.